

RAHUL

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IIT (ISM) Dhanbad

Education

Indian Institute of Technology (ISM) Dhanbad

M.Tech. in Fuel and Energy Engineering (CGPA: 8.71 / 10.0)

Expected May 2027

Dhanbad, Jharkhand

- **Key Coursework:** Coal and Mineral Beneficiation, Biofuels, Fuel Technology, Computational Fluid Dynamics, Powerplant Engineering, Alternate Energy System, Energy Technology, Processing of Liquid and Gaseous Fuels, Mineral Processing Economics and Project Management, Research Methodology

Government College of Engineering Salem

B.E. in Mechanical Engineering (CGPA: 8.27 / 10)

September 2023

Salem, Tamil Nadu

- **Key Coursework:** Thermodynamics, Fluid Mechanics, Heat Transfer, Engineering Mechanics, Strength of Material, Theory of Machines, Manufacturing Engineering, Industrial Engineering

Technical Skills

Languages: Python

Tools & Softwares: Ansys Fluent, PowerBI, JupyterLab, MS Excel, AutoCAD, CATIA v5, MS Word, Powerpoint, LabVIEW, LaTeX

Core Competencies: Computational Fluid Dynamics, Data Analysis, Coal Beneficiation, Life Cycle Assessment, Coal and Biomass Characterization, Crude Oil Characterization, Mineral Economics

Experience

Tata Steel Limited

Research Intern

May 2026 – February 2027

Jamshedpur, Jharkhand

- Built CFD models in ANSYS Fluent with Eulerian two-fluid model with KTGF to predict density based separation of a gas-solid fluidized bed based coal separation.
- Performed a grid-independence study with 6 meshes in Excel to select an optimal mesh with less than 2.5% relative error and validated simulation results against published benchmark data using python with deviations as low as 0.4%.
- Developed Python scripts to parse, filter, clean and transform large volumes of ASCII simulation output, performed the downstream data analysis, and generated structured analytical reports.
- Wrote custom UDFs to vary drag models terms and add sinusoidal vibration force terms.

Projects

Hydrodynamics Studies of Magnetically assisted Vibrating Gas Fluidized Bed | *Ansys Fluent, Python, Excel*

Present

- Built 2D transient CFD models in ANSYS Fluent with Eulerian two-fluid model with KTGF to predict density based separation of a novel gas-solid fluidized bed based coal separation method.
- Performed a grid-independence study with 6 meshes in Excel to select an optimal mesh with less than 2.5% relative error and validated simulation results against published benchmark data using python with deviations as low as 0.4%.
- Developed Python scripts to parse, filter, clean and transform large volumes of ASCII simulation output, performed the downstream data analysis, and generated structured analytical reports.
- Wrote custom UDFs to vary drag models terms and add sinusoidal vibration force terms.

Performance Analysis of Comfort Mattress using PCMs | *CATIA v5, Ansys Fluent, LabVIEW, MS Excel, MS Word*

- Designed and simulated a cooling mattress under the environmental conditions of Tamil Nadu.
- With the use of thermocouples, collected temperature data at 12 heat zones on mattress using NI DAQ System and LabVIEW Software.
- Compared results of ordinary mattress and PCM integrated mattress.
- Developed a mattress to provide cooling by 25%.

Achievements

- One of the three students selected as Research Intern at Tata Steel Jamshedpur.

Positions of Responsibility

- **Class Representative:** MTech Fuel & Energy Engineering, IIT (ISM) Dhanbad
- **Student Internship Representative:** MTech Fuel & Energy Engineering, IIT (ISM) Dhanbad
- **Organizer:** Teacher's Day (also did anchoring) and Saraswati Puja
- **Sports & Extracurriculars:** Active participant in badminton and cricket, Winner in Science and Maths Exhibitions.